# Math Class 9th Federal Board Chapter 03 Sets and Relations Full Test

[5 ~ 10]
2 Short Qs

| | | | | |
|---|---|---|---|---|
| **Sets and Functions** | **[SLO: M-09-A-10]:** Describe mathematics as the study of pattern, structure, and relationships. | Summative | U | **Question(s) will be asked in the annual examination** |
| | **[SLO: M-09-A-11]:** Identify sets and apply operations on three sets (Subsets, overlapping sets and disjoint sets), using Venn diagrams | Summative | A | **Question(s) will be asked in the annual examination** |
| | **[SLO: M-09-A-12]: Solve problems on classification and cataloguing by using Venn diagrams for Scenarios involving two sets and three sets. Further application of sets.** | **Formative** | **U** | **Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.** |
| | **[SLO: M-09-A-13]:** Verify and apply properties/laws of union and intersection of three sets through analytical and Venn diagram method. | Summative | A | **Question(s) will be asked in the annual examination** |
| | **[SLO: M-09-A-14]:** Apply concepts from set theory to real world problems (such as in demographic classification, categorizing products in shopping malls and music playlist by genre) Relation. | Summative | A | **Question(s) will be asked in the annual examination** |
| | **[SLO: M-09-A-15]:** Explain product, Binary Relations and its domain and range. | Summative | K | **Question(s) will be asked in the annual examination** |
| | **[SLO: M-09-A-16]:** Recognize that a relation can be represented by table, order pair and graphs. | Summative | A | **Question(s) will be asked in the annual examination** |

**Q.1 Choose the correct option. (15 × 1 = 15 marks)**

1. Range of {(1, 3), (2, 5)} is:

A. {1, 2} B. {3, 5}

C. {1, 3} D. {2, 5}

Binary

{ }

$(x, y)$

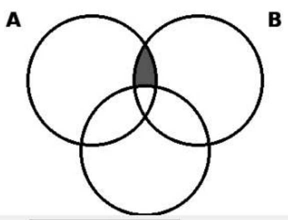
ordered Pair

2. For A = {a, b} and B = {1, 2}, the relation R = {(a, 1), (b, 2)} is a subset of:

A. A × A B. A × B

C. B × A D. B × B

3. In the Venn diagram, the region A ∩ B ∩ C′ represents elements in:




Cartesian product

$A \times B \neq B \times A$

$A \times B = \{(a,1), (a,2), (b,1), (b,2)\}$

1 2 3 4

MCQ.

# of subsets of $A \times B$ =

$n(A \times B) = 4$

$n(A \times B) = n(A) \times n(B)$

$n(A \times B) = n(A) \times n(B)$

$= 2 \times 2 = 4$

$A = \{1, 2\} \Rightarrow 2^2 = 4$

Power set = All subsets of A

$P(A) = \{\phi, \{1\}, \{2\}, \{1,2\}\}$

MCQ No of subsets of A x B

$A = \{a, b\}, B = \{1, 2\}$

a) 4 b) 2, c) 8, d) 16

$n(A \times B) = 2 \times 2 = 4$

# of subsets $= 2^4 = 16$

$A \times B = \{(a,1), (a,2), (b,1), (b,2)\}$

$R \subseteq \underline{A \times B}$ → Cartesian product

↑ Binary Relation

$A = \{a, b\}$, $B = \{1, 2\}$

Table

Carte.

| $A \times B$ | $a$ | $b$ |
|---|---|---|
| 1 | $(a,1)$ | $(b,1)$ |
| 2 | $(a,2)$ | $(b,2)$ |

Ex

$A = \{0, 1, 2\}$, $B = \{3, 4\}$

Q

$\underline{A \times B}$ $n(A \times B) = 6$

# of subset $= 2^6 = 64$

| $A \times B$ | 0 | 1 | 2 |
|---|---|---|---|
| 3 | $(0,3)$ | $(1,3)$ | $(2,3)$ |
| 4 | $(4,0)$ | $(4,1)$ | $(4,2)$ |

$\cap \rightarrow$ common

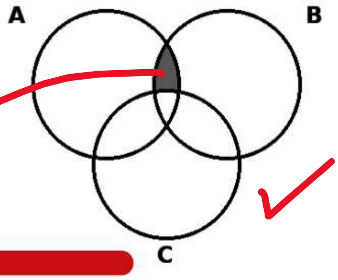
3. In the Venn diagram, the region $A \cap B \cap C'$ represents elements in:

C

$\cup \rightarrow$ OR

$\cap \rightarrow$ AND

$A \cap B \cap C'$




A. A and B but not in C

B. A, B and C

C. A or B but not C

D. A only

D, R

I, II

4. What is the domain of the relation R = {(1, 2), (2, 3), (3, 4)}?

A. {1, 2, 3}

B. {2, 3, 4}

C. {1, 3, 4}

D. {2, 4}

5. Given X = {2, 4, 6}, Y = {1, 3, 5} and R = {(1, 2), (3, 4), (5, 6)}, R is a subset of:

A. X × X

B. X × Y

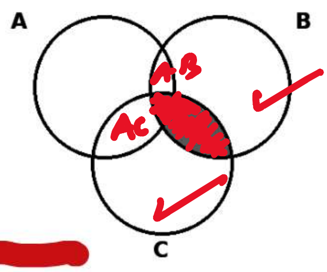
C. Y × X

D. Y × Y

$R = \{(1,1), (3,1), (1,5)\}$

$R \subset Y \times X$

6. What statement does the shaded region in the Venn diagram represent?




A. A ∩ B ∩ C

B. (A ∩ B) ∩ C

C. B ∩ C

D. A ∪ C

7. If A = {2, 4, 6} and B = {0, 1}, then the number of elements in A × B is:

A. 5
B. 6
C. 8
D. 9

$3 \times 2 = 6$

$n(A) \times n(A)$

8. If n(A) = p, then n(A × A) =

A. p
B. 2p
C. $2p^2$
D. $p^2$

$P \times P = P^2$

9. If R = {(a, b) | a, b ∈ N and a × b = 12}, then the tabular form of R is:

A. {(1,12),(2,6),(3,4),(4,3),(6,2),(12,1)}
B. {(1,12),(2,6),(3,4)}
C. {(12,1),(6,2),(4,3)}
D. {(1,12),(4,3),(2,6)}

{( ),( ,}

10. If R = {(a, b) | a, b ∈ Z and a + b = 0} then:

A. Dom R ⊂ Range R
B. Range R ⊂ Dom R
C. Dom R = Range R
D. None of these

11. The number of different ways commonly used to describe a set is:

A. 1
B. 2
C. 3
D. 4

1. Descriptive
2. Tabular
3- set Builder Notation

$N = \{1, 2, 3, 4, 5, 6 \ldots\}$

$(a,b) \rightarrow N \times N$

$A \times B$

$\{1, 2, 3, 4, 5, 6 \ldots 12\} \times \{1, 2, 3, 4 \ldots 6 \ldots 12 \ldots\}$

$\{(1,12), (2,6), (3,4), (4,3), (6,2), (12,1)\}$

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, +1, +2, +3 \ldots\}$$

$a + b = 0$

$\Rightarrow a = -b$

$(1, -1)$, $(2, -2)$, $(-3, 3)$, $(-4, 4)$, $(0, 0)$

$b = -1, \; a = 1$

$b = -2, \; a = -2$

Whole No $= \{0, 1, 2 \ldots\}$

12. The set {x | x ∈ W and x ≤ 101} is:

A. Infinite set

B. Subset

C. Null set

D. Finite set

13. The set-builder form of A - B is:

A. {x | x ∈ A}

B. {x | x ∈ A and x ∉ B}

C. {x | x ∈ A and x ∈ B}

D. {x | x ∈ B}

14. If A ∪ B = A and A ∩ B = B, then:

A. A ⊆ B

B. A ⊉ B

C. A ⊇ B

D. A ≠ B

$A = \{1, 2, 3, 4\}$ $B = \{3, 4\}$

15. If A ⊂ B, then A - B =

$A = \{1, 2\}$, $B = \{1, 2, 3, 4\}$

A. A

B. B

C. ∅

D. B - A

$$W = \{0, 1, 2, \ldots, 101\}$$

$W = \{0, 1, 2, \ldots, 10\}$

$x \in W$

$n(A) = 102$

$A - B \rightarrow$ A difference

$A = \{1, 2, 3\}$ $B = \{\cancel{3}, 4, 5\}$. B

$A - B = \{1, 2\}$ A

$A \subset B \rightarrow$ A is proper subset of B

proper subset superset

$A \subseteq B$

(A is improper subset)

A is a subset.

$A \subseteq B \rightarrow$ A is a subset of B

$A \supseteq B$ A is superset of B

**Q.2 Answer all of the following short questions. (15 × 4 = 60 marks)**

(i) If A = {a, b, c, d}, B = {a, c, e} and U = {a, b, c, d, e, f}, show using a Venn diagram that (A ∪ B)′ = A′ ∩ B′.

2 marks      2 marks

$(A \cup B)' = A' \cap B' \rightarrow$ DeMorgan's Law

$A \cup B = \{a, b, c, d, e\}$

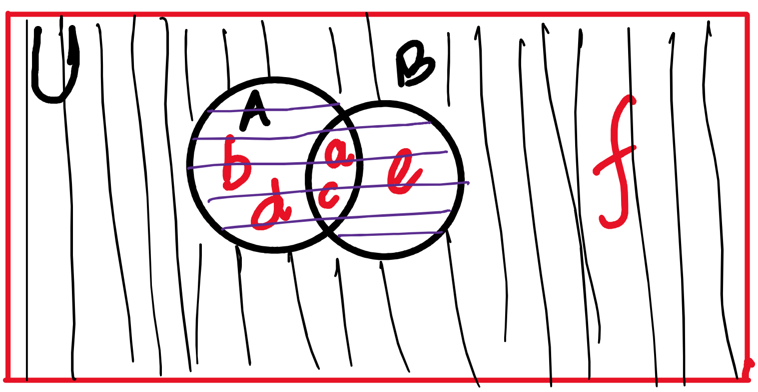
$(A \cup B)' = U - (A \cup B) = \{f\}$ LHS

$A' = U - A = \{e, f\}$

$B' = U - B = \{b, d, f\}$

Sol:- $A' \cap B' = \{f\}$ R.H.S =

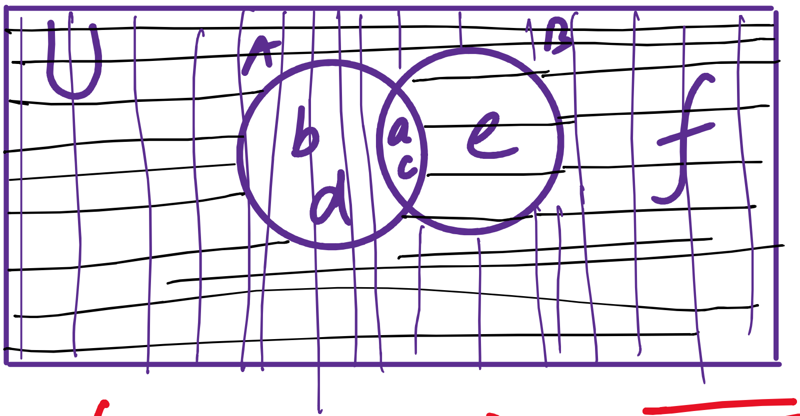
L.H.S.



$A \cup B \longrightarrow \equiv$

$(A \cup B)' \longrightarrow |||$ ✓

R.H.S.




$A' \longrightarrow \equiv$

$B' \longrightarrow |||$

$A' \cap B' \longrightarrow \equiv, |||$

Thus $L.H.S = R.H.S$

$(x, y)$

Let A = {1, 4, 8} and B = {1, 6}. Find the Cartesian product A × B. Also find R = {(x, y) | x ∈ A, y ∈ B and x ≥ y}.

Sol.

a) $A \times B = \{(1,1), (1,6), (4,1), (4,6), (8,1), (8,6)\}$ 2

b) $R = \{(1,1)\ (4,1)(8,1)(8,6)\}$ 2

b) $R = \{(1,1),(4,1)(8,1)(8,6)\}$

If A = {1, 3} and B = {2, 4}:
(a) Write the Cartesian product A × B.
(b) Find $R_1$ = {(x, y) | x ∈ A, y ∈ B and x > y}.

a) $A \times B = \{(1,2),(1,4),(3,2),(3,4)\}$

b) $R_1 = \{(3,2),\}$

(vi) For A = {1, 2, 3} and B = {3, 4}:
(a) List all ordered pairs of A × B.
(b) List all ordered pairs of R = {(x, y) | x ∈ A, y ∈ B and x < y}.
(c) Find the domain and range of R.

a) $A \times B = \{(1,3),(1,4),(2,3),(2,4),(3,3),(3,4)\}$

b) $R = \{(1,3),(1,4),(2,3),(2,4),(3,4)\}$

c) Dom $R = \{1, 2, 3\}$

Range $R = \{3, 4\}$

(xii) For A = {1, 3, 5} and B = {4, 5}:
(a) Find A × B

(xii) For $A = \{1, 3, 5\}$ and $B = \{4, 5\}$:

(a) Find $A \times B$.

(b) List all ordered pairs of $R = \{(x, y) \mid x \in A, y \in B \text{ and } x + y \geq 7\}$.

a) $A \times B = \{(1,4),(1,5),(3,4),(3,5),(5,4),(5,5)\}$

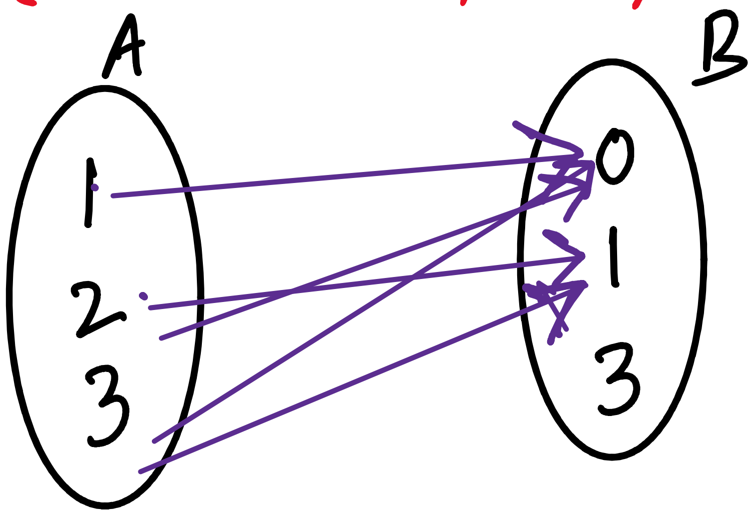
b) $R = \{(3,4),(3,5),(5,4),(5,5)\}$

Let $A = \{1, 2, 3\}$, $B = \{0, 1, 3\}$ and $R = \{(a, b) \mid a \in A, b \in B \text{ and } a > b\}$. Find R and represent it by an arrow diagram.

Sol

$R = \{(1,0),(2,0),(2,1),(3,0),(3,1)\}$

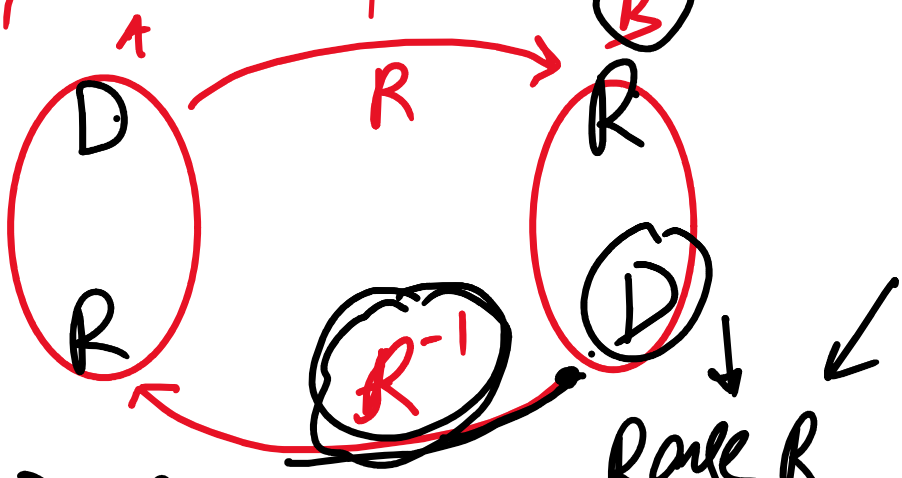
Arrow diagram




(xiv) Given $A = \{0, 2, 3\}$, $B = \{0, 2, 4, 6, 9, 16\}$ and $R = \{(x, y) \mid x \in A, y \in B \text{ and } x^2 = y\}$, verify that $Dom(R^{-1}) = Range(R)$ and $Range(R^{-1}) = Dom(R)$.

Inverse Relation

$$R^{-1} = \{(y,x) \mid y \in B, x \in A \text{ and } x = y\}$$




$R : A \to B$

Dom R

Rang $R^{-1}$

Range R

Dom $R^{-1}$

$$\boxed{\begin{array}{l} \text{Dom } R = \text{Rang } R^{-1} \\ \text{Range } R = \text{Dum } R^{-1} \end{array}}$$

(16)

(9, 4)

(xiv) Given A = {0, 2, 3}, B = {0, 2, 4, 6, 9, 16} and R = {(x, y) | x ∈ A, y ∈ B and $x^2 = y$}, verify that Dom($R^{-1}$) = Range(R) and Range($R^{-1}$) = Dom(R).

$y^2 = x$

Sol.

$\Rightarrow R = \{(0,0), (2,4), (3,9)\}$

$\text{Dom } R = \{0, 2, 3\}$

$Dom\ R = \{0, 2, 3\}$

$Range\ R = \{0, 4, 9\}$

$R^{-1} = \{(0,0), (4,2), (9,3)\}$

$Dom\ R^{-1} = \{0, 4, 9\}$

$Range\ R^{-1} = \{0, 2, 3\}$

Thus $Dom\ R = Range\ R^{-1}$

& $Range\ R = Dom\ R^{-1}$ proved.

Let A = {0, 1, 3} and B = {1, 2, 3, 5, 7}. Write R = {(x, y) | x ∈ A, y ∈ B and y = 2x + 1} in tabular form. Also find $R^{-1}$.

Sol

$x, y$

$R = \{(0, 1), (1, 3), (3, 7)\}$

$x = 0,\ y = 2(0)+1 = 1$

$x = 1,\ y = 2(1)+1 = 3$

$x = 3,\ y = 2(3)+1$

$R^{-1} = \{(1,0), (3,1), (7,3)\}$